

RUT Stage A Preliminary Results

Technical Report for GitHub / OSF - Updated with Stage A.11 dG-Bound Sensitivity

Status

Preliminary background-only result, updated after Stage A.11 dG-bound sensitivity. Scope: BAO observables, cosmic chronometer $H(z)$, H_0 prior, standard-rd and free-rd treatments, diagonal-error testing. This report should be archived with the A5-A11 CSV/JSON outputs and treated as a reproducibility summary, not as a proof claim.

Executive Finding

Stage A now has a clear result: the strongest RUT candidate is not the enlarged-rd route and not the wobble route. The strongest candidate is M1a, a minimal fixed late-time bridge with $z_t = 0.35$, width = 0.03, and one free amplitude parameter dG. In the combined $H(z)$ +BAO channel, M1a is BIC-favored over LCDM in both standard-rd and free-rd scenarios.

The result is specific: it lives mainly in transverse BAO distance, D_M/r_d . It does not win in $H(z)$ -only, and D_H/r_d -only is inconclusive. Therefore Stage A should be described as a low-redshift transverse BAO-distance signal, not a broad background-expansion victory.

Key Pre-A11 Findings

A5 identified the minimal bridge as stronger than the full flexible bridge. A6 showed the fixed-grid result was not a one-point bridge-location artifact. A7 and A8 showed the signal is sensitive to the BOSS Galaxy $z = 0.35$ D_M/r_d observable. A9 showed the improvement comes from BAO residuals while $H(z)$ is slightly worsened. A10 confirmed the channel split: DM-only, BAO-all-only, and combined $H(z)$ +BAO favor M1a; $H(z)$ -only favors LCDM; DH-only is tied or inconclusive.

Stage A.11: dG-Bound Sensitivity

A11 tested whether the M1a result requires extreme negative dG values by repeating the fits under dG bounds of ± 0.10 , ± 0.15 , ± 0.20 , and ± 0.25 . This is the final Stage A hardening check before moving to Stage B.

Scenario / Channel	dG ± 0.10	dG ± 0.15	dG ± 0.20	dG ± 0.25	Read
Standard rd + H_0 / DM only	-7.125	-10.515	-12.976	-14.291	Strong; hits lower bound
Standard rd + H_0 / BAO all	-6.299	-8.972	-10.211	-10.258	Strong; nearly saturates
Standard rd + H_0 / $H(z)$ +BAO	-5.180	-6.778	-6.822	-6.822	Strong; stabilizes near dG = -0.159
Free rd + H_0 / DM only	-2.358	-3.502	-4.755	-6.128	Moderate to strong; hits lower bound
Free rd + H_0 / BAO all	-3.273	-5.577	-7.395	-8.512	Moderate to strong; hits lower bound
Free rd + H_0 / $H(z)$ +BAO	-4.628	-6.084	-6.118	-6.118	Strong; stabilizes near dG = -0.159

Interpretation of A11

The combined $H(z)$ +BAO result does not require the maximum allowed correction. In both standard-rd and free-rd treatments, the result is already favored at dG ± 0.10 , becomes strong around dG ± 0.15 , and saturates once the best-fit value dG is allowed to reach approximately -0.159. The ± 0.20 and ± 0.25 combined fits

produce essentially the same BIC because the optimum is interior, not pinned to -0.25.

The DM-only and BAO-only channels still prefer stronger negative dG and often hit the lower bound. That is expected from the A10 diagnosis: transverse BAO wants a stronger bend, while $H(z)$ acts as a brake in the combined test. The combined channel is therefore the cleaner result because it balances BAO improvement against expansion-rate pressure.

Updated Reviewer-Safe Claim

In preliminary diagonal-error Stage A testing, a one-parameter fixed late-time bridge centered near $z_t = 0.35$ improves low-redshift transverse BAO-distance residuals enough to outperform LCDM by BIC in BAO and combined $H(z)$ +BAO channels. A11 shows the combined-channel advantage does not require extreme dG ; it stabilizes near $dG = -0.159$. The signal is strongest in D_M/r_d , not $H(z)$ -only or D_H/r_d -only.

Numerical Snapshot

Quantity	Best combined standard-rd	Best combined free-rd	Meaning
z_t	0.35 fixed	0.35 fixed	Bridge center near low-redshift BAO anchor
width	0.03 fixed	0.03 fixed	Sharp late-time transition
dG	about -0.1594	about -0.1585	Moderate bridge amplitude
rd	about 146.77	about 146.63	Standard-ish ruler, not enlarged-rd
H_0	about 73.64	about 73.66	Near local- H_0 side
Delta BIC	about -6.82	about -6.12	Strong preliminary BIC support

Limitations and Next Required Tests

- The result is preliminary and diagonal-error only; covariance-aware BAO must be tested.
- The decisive Stage A signal is tied to BOSS Galaxy $z = 0.35$ D_M/r_d and must survive updated BAO compilations.
- $H(z)$ -only favors LCDM, so this is not yet a general expansion-rate victory.
- DM-only and BAO-only keep asking for more negative dG , so the combined channel should be emphasized over isolated DM-only claims.
- Stage B growth/lensing and Stage C joint consistency remain mandatory before paper-facing claims.

Recommendation

Freeze Stage A as a preliminary background-lane result and move to Stage B. Upload this updated report, the A5-A11 result tables, plots, and JSON summaries to GitHub/OSF. Then test whether the bridge remains healthy under growth, lensing, $f_{\sigma 8}$, $\mu(k,a)$, $\Sigma(k,a)$, and E_G -style diagnostics.